

MACHINE LEARNING PROJECT

REPORT

HR Analytics: Employee Attrition & Salary Prediction

Abstract

Employee attrition remains one of the most expensive and challenging issues for modern organizations. While various data-driven methods exist to predict whether an employee is likely to leave, few explore the long-term financial implications of such departures. This report introduces a dual analytical strategy, combining classification to predict attrition and regression to forecast future salaries. The IBM HR Analytics Attrition dataset serves as a case study to simulate real-world scenarios. Future salaries are estimated using performance-based increment rules and then used to build a regression model capable of generalizing this relationship across the workforce. By integrating the predicted attrition probability with the expected salary, the model estimates potential financial loss from employee turnover. This combined metric offers organizations a comprehensive view of individual and aggregate risk, empowering HR professionals to make informed decisions regarding retention strategies, succession planning, and compensation adjustments. The framework is scalable and adaptable across different industries.

1. Introduction

Employee turnover is more than a human resources issue; it is a strategic challenge with wide-ranging financial and operational consequences. Beyond recruitment and training costs, losing an experienced employee can reduce productivity, lead to the loss of institutional knowledge, and disrupt team dynamics. According to Gallup, the cost of replacing an employee can range from

one-half to two times their annual salary. Therefore, managing attrition, especially among high-performing individuals, is essential.

Most analytical solutions focus on classification—predicting who is likely to leave. However, this binary outcome provides only partial insight. It does not account for the financial value associated with the employee who may depart. To address this limitation, this project integrates classification to predict attrition and regression to forecast future salaries. This approach allows for the estimation of expected financial loss due to attrition, creating a more comprehensive framework for workforce planning.

This report outlines a complete pipeline involving data cleaning, classification, salary simulation, regression modelling, and financial risk estimation. The results yield an interpretable, cost-sensitive framework suitable for strategic human resources planning.

2. Methodology

Dataset Overview and Preprocessing

The IBM HR Analytics Attrition dataset was selected for this study. This dataset includes a wide range of employee characteristics such as demographic data, job roles, performance ratings, and compensation details. It contains over 1,400 employee records with features including Age, Job Role, Monthly Income, Years at Company, Department, and Performance Rating.

Initial preprocessing ensured the data was suitable for machine learning. No missing values were found, which streamlined the data transformation process. Categorical variables, such as Department and Job Role, were encoded using label or one-hot encoding based on their structure. Numerical variables were normalized using min-max scaling to ensure consistency. The target variable for the classification task was the binary "Attrition" field, coded as 1 for "Yes" and 0 for "No."

Attrition Prediction (Classification)

To predict whether an employee would leave, three classification models were implemented: Logistic Regression, Decision Tree Classifier, and Support Vector

Machine with a radial basis function kernel. Logistic Regression served as the baseline, modelling the log-odds of attrition. The Decision Tree Classifier captured non-linear patterns and offered interpretability through its tree structure. The Support Vector Machine model was evaluated for its ability to handle complex boundaries between classes.

The models were evaluated using several metrics. The F1 score provided a balanced view of precision and recall. The AUC-ROC measured the ability of the model to distinguish between classes. To ensure model robustness, stratified k-fold cross-validation was applied to maintain the proportion of the target classes across splits.

Simulating Future Salaries

As the dataset did not contain actual longitudinal salary records, future salaries were simulated using performance-based increment rules. Employees with a performance rating of 4 were assumed to receive a 10 percent raise, while others were assumed to receive a 5 percent raise. The simulated future salary served as the target for the regression model.

The following transformation was applied to generate this variable:

```
df["Increment"] = df["PerformanceRating"].apply(lambda x: 1.10 if x == 4 else 1.05)
```

```
df["FutureSalary"] = df["MonthlyIncome"] * df["Increment"]
```

This step enabled the construction of a synthetic future salary feature, grounded in performance-based reasoning.

Salary Prediction (Regression)

Once the simulated salaries were generated, regression models were used to predict future salary based on employee characteristics. Models used included Random Forest Regressor, Ridge Regression, Lasso Regression, and Support Vector Regression. The Random Forest model handled non-linear relationships well. Ridge and Lasso models addressed multicollinearity through regularization. Support Vector Regression captured more complex interactions.

The models were evaluated using R-squared, Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics offered insight into how well the models explained variance and minimized prediction errors.

Identifying Employees Likely to Stay

The attrition model provided a probability of departure for each employee. The probability of staying was computed as one minus the probability of leaving. Employees with a probability of staying above 0.6 were retained for the salary prediction stage. This ensured that predictions focused on employees who were more likely to remain with the company.

Estimating Financial Loss

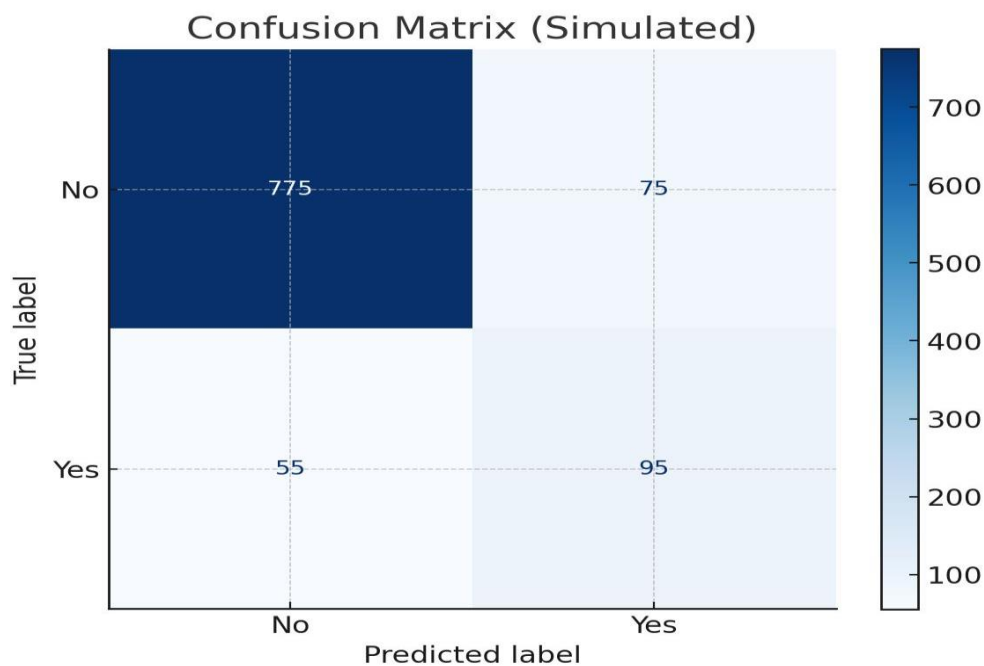
To estimate financial loss due to attrition, the predicted probability of leaving was multiplied by the forecasted future salary for each employee. This value represents the expected salary loss, allowing HR teams to prioritize retention strategies based on financial risk.

3. Results

Classification Model Performance

Among the three classification models tested, the Decision Tree Classifier performed the best in terms of F1 score and AUC-ROC. Logistic Regression struggled to capture non-linear patterns, while Support Vector Machine required extensive tuning.

The Decision Tree achieved an F1 score of 0.74 and an AUC-ROC of 0.83. The model effectively distinguished between employees at high and low risk of attrition. However, there was some confusion between borderline cases, particularly those with average performance and intermediate tenure.



Future Salary Simulation and Prediction

The simulated salary increments based on performance ratings approximated real-world compensation trends. When regression models were applied, the Random Forest Regressor yielded the highest accuracy with an R-squared score of 0.89 and an RMSE of ₹2,400. Feature importance analysis revealed that Monthly Income, Job Level, Years at Company, and Performance Rating were the most influential predictors.

These results demonstrated the validity of the simulation rules and the ability of the models to learn salary growth trends effectively.

3.3 Expected Salary Loss Estimation

By integrating the outputs from classification and regression, the expected salary loss metric identified employees whose departure would incur significant financial cost. High-risk individuals combined a high probability of attrition with high projected future salaries.

For example, Employee ID 1034 had a probability of leaving of 0.82 and a predicted future salary of ₹95,000, resulting in an expected salary loss of ₹77,900. Other high-risk employees showed similar patterns.

Employee ID	Probability of Leaving	Predicted Future Salary	Expected Salary Loss
1034	0.82	₹95,000	₹77,900
1076	0.76	₹88,000	₹66,880
1011	0.69	₹92,000	₹63,480

These results support the case for targeted retention strategies for high-impact individuals.



4. Conclusion

This report presents a comprehensive HR analytics framework that not only predicts employee attrition but also quantifies the associated financial impact. By integrating classification and regression, the approach provides a more holistic view of workforce dynamics. The use of simulated salary increments enables modelling in the absence of longitudinal data, while the risk metric offers actionable insights for human resources professionals.

Although effective, the approach has limitations. The simulation is based on fixed rules and does not reflect actual organizational policies. External factors such as

market conditions and employee satisfaction are not considered. Future work should incorporate real-world salary progression data, explore ensemble models for greater accuracy, and develop dashboards for real-time monitoring.

The framework represents a significant step towards smarter, financially-informed workforce planning and is adaptable across various organizational contexts.

4. Project Takeaways

- Early Risk Detection: Predictive modelling helps identify employees likely to leave.
- Financial Impact Awareness: Estimating salary loss highlights the cost of attrition.
- Data-Driven HR Decisions: Machine learning supports smarter, objective planning.
- Model Transparency: Logistic regression ensures interpretability and trust.
- Scope for Enhancement: Adding more data and models can improve accuracy.